Marvin Koss

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Research Experience

Accrue (accrue.build) — Independent Research Feb 2026 – Apr 2026
FR8 *Cohort 2.s, Helsinki*

- Reimplemented and extended SLAO (arXiv:2512.23017) with Elastic Weight Consolidation. Achieved +1% BWT with Llama-2-7B-Chat on the Long Sequence Benchmark vs. previous SotA of −2.9% BWT.
- Built an LLM agent that learns across sessions via *weight-level* continual learning without RAG or context management.

Section Cognitive Neuropsychiatry, UK Heidelberg — Research Assistant Aug 2025 – Jan 2026
Supervisor: Dr. Mike Schmitgen

- Built a preprocessing pipeline for fMRI data on 30 subjects in aging study; used by the group for downstream analysis.

StatNLP Group, Heidelberg — Research Assistant Nov 2024 – May 2025
Supervisor: Prof. Dr. Stefan Riezler

- Mathematically proved model collapse for ML models trained on deterministic decision rules in sepsis data; verified with empirical data.

IWR, Heidelberg — Research Assistant Jan 2023 – Sep 2024
Supervisors: Prof. Dr. Jakob Zech, Prof. Dr. Simon Weißmann

- Mean-field analysis of consensus-based optimization: compactness argument for solubility of the associated Fokker–Planck equation; resulted in publication (see Publications).
- Designed and implemented a novel particle-based optimizer with improved exploration over CBO; code.

ZITI, Heidelberg — Research Assistant Nov 2020 – Oct 2022
DeepChip project: NN quantization & uncertainty quantification

- Built a PyTorch framework for sub-INT8 quantization of arbitrary NNs; used it to fully quantize one-block transformers to $<$ 8-bit at low accuracy loss.
- Developed a UQ tool for Bayesian neural networks (slides).

Publications

Marvin Koss, Simon Weissmann and Jakob Zech, *On the Mean-Field Limit of Consensus-Based Methods* (2026), *Mathematical Methods in the Applied Sciences*, 49(5), 4214-4240 10.1002/mma.70343.

Education

Heidelberg University 2024 – Present
M.Sc. Scientific Computing (GPA 1.7 (German)) — in progress

Heidelberg University 2022 – 2024
B.Sc. Mathematics (GPA 2.1 (German))

Heidelberg University 2017 – 2024
B.A. Computational Linguistics (GPA 1.3 (German))

Selected Projects

Galvani 2026
A Connectome to Neural Dynamics Pipeline — code

SimoSS 2022
3D Optimiser visualisation in C++ — code

Teaching

TA, *Formal and Computational Semantics*, Heidelberg, WS 2018 — ran tutorials, designed Python assignments, gave some lectures.

TA, *Formal Mathematical Foundations for Computational Linguists*, Heidelberg, SS 2018.

Skills

Programming: Python (9 years; PyTorch, NumPy, Transformers, Pyro, scikit-learn); C++

ML: transformers, continual learning, PEFT, optimization (theory; gradient & particle-based), mechanistic interpretability.

Math: probability theory, SDEs, mean-field analysis, optimal transport.

Languages: German (native), English (fluent, C1), Russian (basic).